

Business Summary: Performance-Based System Development and IT Investment Assessment

March 24, 2022 — Hirano System Design Office

Business Overview

We are considering launching an IT Assessment Consulting and IT Assessment Tool business using methods developed to quantify and evaluate the effects of performance-based system development and IT utilization. I am Naohiko Hirano of Hirano System Design Office.

Personal Background

After joining CSK (now SCSK) and subsequently becoming an independent professional, I have worked with domestic manufacturers, system integrators, software houses, and user organizations. My work has included business planning using IT; planning and implementing financial-sector businesses such as credit, non-bank finance, securities, and transportation-related electronic money; quality improvement and standardization in project implementation; outsourcing business planning and management; and establishment of internal controls. I have also worked as a part-time instructor in system development at a vocational school.

Since 1990, I have studied IT architecture and related methodologies, followed by research into methods for evaluating IT investment. I have been involved with computers for approximately half a century and have conducted research for approximately 30 years.

Background

IT is a value-added industry, but much of its financial structure is based on a man-month calculation and a multi-tier subcontracting structure. Under a man-month model, revenue is proportional to the number of employees and their utilization rate. If employee utilization declines significantly, it becomes difficult to sustain the business. This issue is expected to become increasingly serious because of Japan's declining birthrate and the large-scale retirement of the post-baby-boom generation.

The multi-tier subcontracting structure itself is not necessarily bad. However, as work moves from the prime contractor to first-, second-, and third-tier subcontractors, the unit price decreases. In extreme cases, the amount reaching the lowest tier can be less than half of the amount paid by the prime contractor. This issue was also pointed out in the Fair Trade Commission's 2022 survey report concerning subcontracting transactions in the software industry.

Naturally, technical levels may also decline. Although reskilling is being advocated, the structure does not necessarily provide sufficient motivation for learning and improving technical capabilities. This point was also discussed in the Information-technology Promotion Agency, Japan (IPA) DX White Paper 2023: "Digital" Getting Underway, "Transformation" Not Progressing.

Another current condition is that companies with 300 or fewer employees account for approximately 70% of establishments engaged in contract-based IT business. Sales are concentrated almost entirely in Japan, and the number of companies and personnel capable of conducting business overseas is limited. Japan can no longer afford to overlook the possibility that it is lagging behind internationally in IT. This situation inhibits progress both in the IT industry and in the use of IT by organizations, and may also be one factor contributing to stagnation in Japanese industry.

Therefore, in order to move away from the existing contract-based, man-month and labor-intensive business model, it is necessary—although this is an ambitious approach—to change the revenue structure. Changing the revenue structure requires a fundamental renewal of the existing contract-based business model. This includes reforming development business processes and contracting methods,

introducing evaluation systems and evaluation tools, substantially improving the means of implementation—such as human resources, methodologies, technologies, tools, and reuse—and reviewing organizational planning, including finance, personnel evaluation, and education.

Especially if the evaluation of IT investment and the provision of evaluation tools are viewed as increasingly important as DX and the use of AI (artificial intelligence) become global trends, this will be an important issue for companies, local governments, national governments, and other countries.

Background — Continued

Corporate management strategies are becoming shorter in cycle: from annual reviews to semiannual reviews, then quarterly reviews, and increasingly to monthly reviews. Without an IT investment evaluation method and tools that can provide timely information and visualize it, organizations may find it difficult not only to advance DX but even to maintain large-scale systems. Although this is an extreme perspective, the possibility of consequences extending to broader social disruption should also be considered.

Companies may also come to need methods for disclosing IT investment evaluation to society, in a manner comparable to the international standard ISO 30414 for human-capital disclosure.

How to Realize the Approach

- **Business processes:** Shift the perspective from one focused on creating things to one focused on achieving results, including profit contribution plans, IT contribution balance analysis, effect-verification plans, investment recovery, benefit-distribution plans, and risk plans for factors that may prevent realization.
- **Contracts:** Apply game theory. Rather than relying on the conventional cost build-up approach, clarify situations in which gains and benefits are exchanged within interdependent relationships and establish a benefit-distribution plan.
- **Evaluation based on the concept of contracts:** In addition to evaluations based on conventional financial and management accounting, introduce innovative accounting. Analyze numerical values for IT and non-IT factors and compare them with cost figures and profit-contribution figures. People, IT, equipment, and other assets are referred to as resources; all activities performed by these resources form the basis for evaluation reporting, analysis, and planning.
- **High-performance organizations:** Aim for high added value, high quality, short development periods, low cost, and, depending on the system, enhanced safety. Improve methods for evaluating and measuring methodologies, tools, and technologies so that they can be promoted effectively. At the same time, develop and implement personnel, education, talent acquisition, financial, and risk plans to create organizations in which people grow, possess autonomy and purpose, communicate skillfully with empathy, and are supported by finance to generate innovation.

Finally

The IPA's DX White Paper 2021 stated that many companies were indifferent to DX or had not achieved results, and that most companies in practice were stalled at the entrance to DX. It also reported that, although a certain number of personnel had the potential to transition to advanced IT roles, motivation and appropriate support to encourage actual transition were insufficient. On the individual side, it found that people were not adequately aware of their relative value in the labor market and lacked confidence in their competitiveness.

The objective is to establish an IT investment evaluation system and shift from a man-month-based model to a performance-based, continuing business model in which the maximum customer value is generated through IT, together with a compensation system commensurate with that value.

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